WILD 591: SEM in Ecology

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Office Hours: Wed. 1100-1200, Fri. 1000-1100
Course website: github.com/thomasriecke/WILD591

Overview

Lecture: 930-1050 Tuesday and Thursday

Laboratory: NA

Prerequisite(s): Undergraduate students should obtain instructor permission prior to enrolling or auditing.

Grading: This course is offered for a traditional letter grade only. It is not offered under the credit or no credit option. Please see Table 2 in this syllabus for a complete breakdown of how your grade will be calculated in this course. Students will be graded based on their performance on 10 homeworks, a project, as well as their participation in the class.

Career Competencies:

- **Communication (CCOM):** Clearly and effectively exchange information, ideas, facts, and perspectives with persons inside and outside of an organization.
- **Critical thinking (CCRT):** Identify and respond to needs based upon an understanding of situational context and logical analysis of relevant information.
- **Technology (CTEC)**: Understand and leverage technologies ethically to enhance efficiencies, complete tasks, and accomplish goals.

Course Objective: The objective of this course is to build familiarity with the use of causal structural equation models for ecological data, and to help you make progress towards completing your dissertation or thesis. This will include qualitative construction of models based on existing evidence in the literature and specific hypotheses developed by the student, as well as quantitative estimation of ‘pathways’ among variables. Throughout the course, we’ll use paradoxical examples from ecology and other fields to clarify the need for structural/causal thinking in statistics when using observational data. Students will gain familiarity with data handling, a number of R packages including ‘lavaan’ and ‘blavaan,’ and constructing their own Bayesian structural equation models. The primary goal of the class is for students to explore the use of SEMs with their own data, but the instructor can provide or simulate data if this is untenable.

Readings: Readings from textbooks, scientific journals and other readings will be posted on the course GitHub at least one week prior to class.

Exams: There are no exams.
Quizzes: There are no quizzes.

Homework: There will be ten homeworks that should be submitted on Canvas. These will be posted weekly, with questions inside R scripts that are discussed in class. Students are free to simply answer inside the R script, or create a new file.

Lecture and lab schedule

Date	Day	Lecture
Section 1		**Linear models and generalized linear models**
25 Aug	Tues	Introduction
27 Aug	Thur	An intro to Bayesian priors and MCMC
1 Sep	Tues	Linear models: lm(), brms(), JAGS, and custom samplers
3 Sep	Thur	Random and fixed effects
8 Sep	Tues	glm() & link functions
10 Sep	Thur	Multicollinearity! Egads!
Section 2		**Structural equation models**
15 Sep	Tues	Model selection
17 Sep	Thur	Understanding direct and indirect effects
22 Sep	Tues	SEMs: lavaan(), blavaan(), and JAGS
24 Sep	Thur	Latent variables
29 Sep	Tues	Latent variables II
1 Oct	Thur	Cross-lags or dynamic SEMs
Section 3		**Case studies & revisiting advanced concepts**
6 Oct	Tues	Goodness-of-fit testing
8 Oct	Thur	Resource selection functions
13 Oct	Tues	Reproductive success and trade-offs
15 Oct	Thur	Occupancy models
20 Oct	Tues	Ordinal regression (Likert scores)
22 Oct	Thur	Ordinal regression (number of offspring)
27 Oct	Tues	Body condition, life-history trade-offs, and individual quality
29 Oct	Thur	Capture-recapture and density-dependence
3 Nov	Tues	Integrated population models
5 Nov	Thur	Reviewing and revisiting concepts
Section 4		**Case studies & revisiting advanced concepts**
10 Nov	Tues	Coding assistance in classroom
12 Nov	Thur	Coding assistance in classroom
17 Nov	Tues	Coding assistance in classroom
19 Nov	Thur	Coding assistance in classroom
24 Nov	Tues	Coding assistance in classroom
26 Nov	Thur	THANKSGIVING (no class)
1 Dec	Tues	Student presentations
3 Dec	Thur	Student presentations

Course Learning Outcomes

1. Manage and format data in R for ecological analyses
2. Construct research questions, hypotheses and predictions
3. Use structural equation models to assess relevant hypotheses given the data
4. Draw evidence-based conclusions and interpret parameter estimates
5. Communicate conclusions for audiences with different levels of understanding (e.g., scientists, managers, and the general public).

Grading

Category	Description	Points
Homework(s)	R script modification, problem solving	100
Project	Summary	10
	Paper	65
	Oral presentation	25
Participation		50
Total		**250**

Project timeline

Date	Assignment
24 Sep	Topic summary
1 & 3 Dec	Oral presentations
10 Dec	Paper due

Additional pertinent information

All members of the University of Montana (UM) community – no matter their backgrounds, hometowns, life experiences, religious beliefs, abilities, or political views – should experience a UM learning and working environment that cultivates belonging and robust intellectual growth. Committed to fostering inclusive prosperity across Montana and beyond, we believe we are a stronger educational institution when we meaningfully include and remain curious about a broad swath of perspectives and experiences.

As a public-serving institution whose student body reflects the-hard working population of Montana and the diverse identities they represent, we believe that building a sense of belonging for all and providing exposure to a range of perspectives is a precondition for the type of learning that prepares our graduates for a complex, pluralistic society.

As members of the University of Montana community, we aspire to:

- Respect the dignity and rights of all persons.
- Practice honesty, trustworthiness, and academic integrity.
- Promote justice, learning, individual success, and service.
- Act as good stewards of institutional resources.
- Respect the natural environment.

Land Acknowledgment: The University of Montana acknowledges that we are in the aboriginal territories of the Salish and Kalispel people. Today, we honor the path they have always shown us in caring for this place for the generations to come.

Student Conduct: Please reference the Student Conduct Code for any questions about student conduct. Briefly, do not use cell phones during class time. Cheating or plagiarism on any assignment or exam will result in a 0. Treat your fellow students in the same way that you wish to be treated. Discrimination or harassment of any type will not be tolerated.

Relevant dates: Please come speak with the instructor as soon as possible if there are any lingering override issues. August 23rd is the last day to drop fallclasses on Cyberbear with no ’W’ (withdrawl) on your transcript and receive any applicable refunds. September 14th is the last day to drop the course with no record on their transcript(those who drop after September 14th will receive a ’W’ on their transcript). December 4th is the last day to drop. Please see this webpage (https://www.umt.edu/withdrawal/deadlines.php) for more information.